Section 1 Exercises and Questions – Embedded Level 2 – ISE – Helwan National University

Exercise 1:

A simple code to demonstrate the capability of the Bitwise operators on two hexadecimal numbers.

Exercise 2:

A simple Code to demonstrate the capability of the Compiler Optimization with Logical operators.

Exercise 3:

Write a C Code to check if the input number is even or odd using If Conditions.

Exercise 4:

Write a C program to determine if a student is eligible to participate in an exam based on the following criteria:

- The student must be in either grade 10 or grade 12.
- The student's age must be between 15 and 18 years (inclusive).
- Additionally, the student must not have previously failed the exam.

Exercise 5:

Write a C Code to take the student grade (A, b, …etc.) from the user and check then print the Equivalent Grade for it (Ex: Excellent, Very Good, …etc.) using Switch Statements (Hint: The Program shouldn't be case sensitive).

Exercise 6:

Develop a program that takes a month number (1-12) as input and prints the number of days in this month using switch statement.

Exercise 7:

Write a C Code to calculate and print the factorial of any input number using loops.

Exercise 8:

Write a program to display a full pyramid using stars pattern taking the number of rows from the user as input.

Ex if User Enters Number of Rows = 5:

```
    *
   ***
  *****
 *******
*********
```

Exercise 9:

Write a C Program to reverse a given multidigit integer using a while loop.

(Ex: Input 235, output 532).

Exercise 10:

Develop a Program that Continuously prompt the user for a calculation until they choose to exit (Hint: Use the Do-While Loop)

Example:

Enter an operator (+, -, *, /):

Enter two operands: num1 and num2

Do you want to perform another calculation? (y/n):

Exercise 11:

Write a C Function that takes one character and checks if it is alphabetic or not.

Exercise 12:

Write a C Function that takes two numbers and checks for the minimum value of them and return it.

Exercise 13:

Write a C Function to calculate the factorial of any input number using Recursion.

Exercise 14:

Write a C Function that calculate the Fibonacci series using recursive method.

The Fibonacci Series: 0,1,1,2,3,5,8,13, 21...

Exercise 15:

A Simple Example to demonstrate the difference between local and global variables in C Programs.

Exercise 16:

Write a C function that calculates the required heater activation time according to the input temperature of water.

- if input temperature is from 0 to 30, then required heating time = 7 mins.
- if input temperature is from 30 to 60, then required heating time = 5 mins.
- if input temperature is from 60 to 90, then required heating time = 3 mins.
- if input temperature is more than 90, then required heating time = 1 mins.
- if temperature is invalid (more than 100), return 0

Example: Input = 10 --> output = 7, Input = 35 --> output = 5

Exercise 17:

Write a program to print the square for numbers from 1 to 10.

Exercise 18:

Write C Function that converts any letter from lower-case to upper-case.

Exercise 19:

Write a C Program for a simple calculator that takes 2 numbers and print the addition, subtraction, multiplication and division using Functions.

Exercise 20:

Write a C Function that displays Prime Numbers between Intervals (two numbers) by Making Function.

Thank You
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